1 Introduction

The objective of this laboratory experiment is to explore the fundamentals of first-order circuits and their practical applications. This includes gaining an understanding of the behavior of capacitors and resistors in circuits and observing the operation of a first-order circuit as an oscillator using the LM555 integrated circuit.

2 Preliminary Work

2.1 Task 1

In the preliminary work section, we are presented with a circuit to analyze, involving a resistor (R) and a capacitor (C). By calculating the capacitor voltage as a function of time for a 5V step input, we aim to determine the behavior of the capacitor in response to this input. The circuit schematic is provided in Figure 1.

1. We have analyzed the circuit shown in Figure 1. Given values are R = 1k ohm and C = 200n F

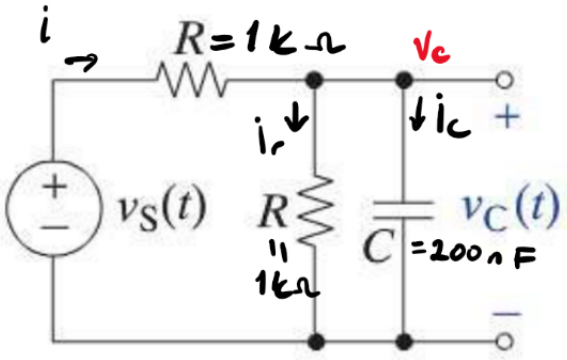


Figure 1: First figure to analyze.

2. We have main current which is (i) is divided to 2 i's they are shown in figure 1.

3. We can calculate the i as
   $i = i_r + i_c -> i = (V_s - V_c)/R$
   $i_r = V_c/R$ and $i_c = C(d/dt)(V_c)$

   if we sum up i_r and i_c we can get

   $V_c = Ae^{(2t/(RC))} + B$, where A and B is stated as constants.

4. We have to determine constants A and B based on initial and infinite conditions. As time approaches infinity, the capacitor acts as an open circuit, causing Vc to approach 2.5V. When t tends to infinity, the expression $Ae^{(2t/(RC))} + B$ simplifies to B. Hence, **B equals 2.5V**. Moreover, we know that V_c is zero at t = 0+. Consequently, **A is equal to -2.5V.**
   That shows us $Vc = 2.5(1 - e^{-10^4 t})$

2.2 Task 2

We need to understand the LM555 timer circuit by studying its data sheet which given in Figure 2. Then, we'll build a circuit using the LM555 timer to create a blinking LED with a 1-second cycle time, where the LED is on for 0.5 seconds and off for 0.5 seconds. To do this, we'll search for a suitable application circuit that generates square wave oscillations using the LM555. The oscillation frequency will be determined by choosing appropriate resistor (R) and capacitor (C) values, and we'll calculate the bias resistor for the LED based on the LED data sheet.

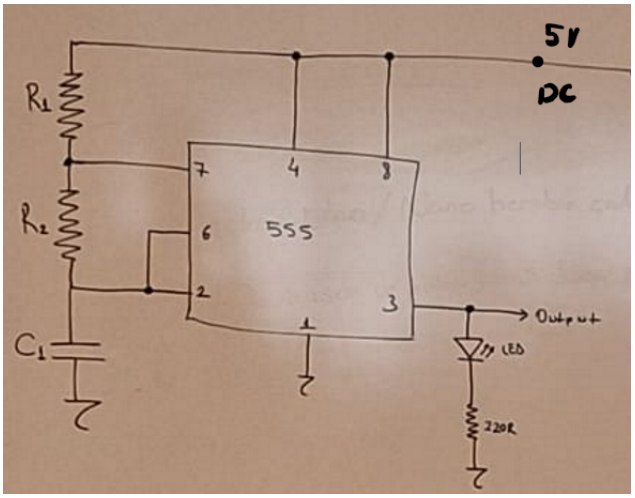


Figure 2: Schematic of LM555 timer circuit.

1. Equations of that circuit is
   $f = 1.44/((R_1 + 2R_2)C)$ and $D = (R_2)/(R_1 + 2R_2)$

   To build a circuit using the LM555 timer IC for a continuous LED blinking with a 1-second cycle 0.5 seconds on and 0.5 seconds off frequency must be 1.

3 Lab Tasks

The aim of the tasks is to apply the concepts of first-order circuits and the LM555 timer IC in an oscillator circuit

3.1 Task 1

1. We constructed the RC circuit that was analyzed in the preliminary work as shown in figure 3.

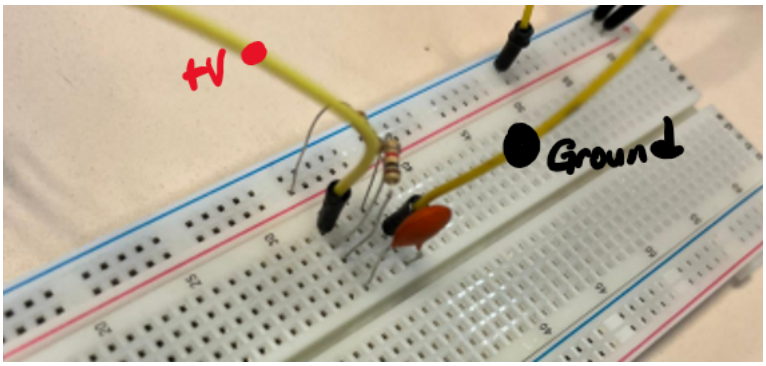


Figure 3: Task 1 completed circuit.

2. After constructing the circuit, I set the voltage to 5 V and the input signal to a step signal. The half period must be 10 times the time constant were also a given. For this circuit, the time constant is equal to RC/2; the resistor values is 1k ohm and capacitor is 220nF. Time constant is therefore equal to $1.1 * 10^{-4}$, which implies that period is $20 * timeconstant = 20 * 1.1 * 10^{-4} = 2.2 * 10^{-3}$. Additionally, since f*T = 1 and f = 103/2.2 = 455Hz, so we set the function generator to 455Hz as shown in figure 4

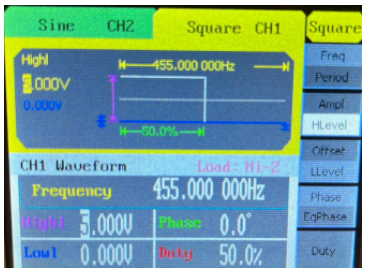


Figure 4: Function generator of task 1 with 455Hz in square wave type.

3. Then we connected inputs and outputs to oscilloscope to see graphs of signals over the circuit.

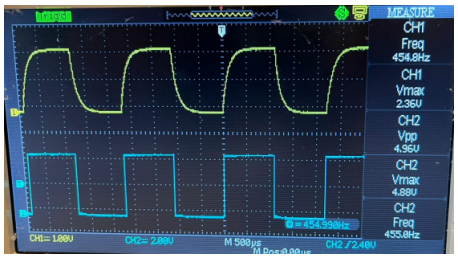


Figure 5: Oscilloscope of task 1 with 455Hz in square wave type.

4. Then we changed frequency to 1KHz.

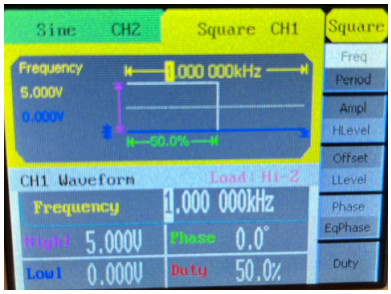


Figure 6: Function generator of task 1 with 1KHz in square wave type.

5. Observed the input - output relation on oscilloscope again as we did in step 3.

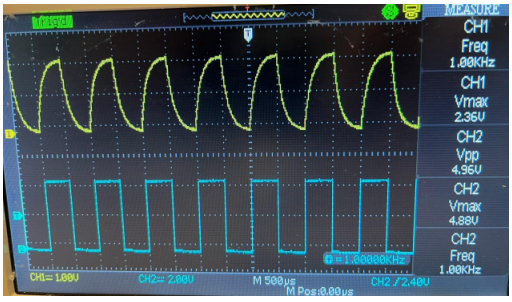


Figure 7: Oscilloscope of task 1 with 1KHz in square wave type.

6. Then we set the frequency to 10KHz.

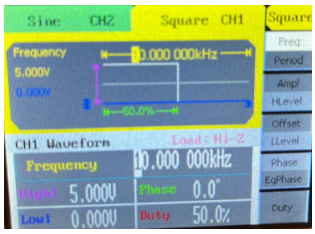


Figure 8: Oscilloscope of task 1 with 10KHz in square wave type.

7. Observed the input - output relation on oscilloscope again as we did in step 3.

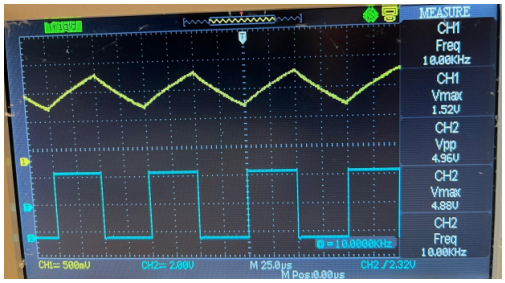


Figure 9: Oscilloscope of task 1 with 10KHz in square wave type.

As the frequency increased, the time available for the capacitor to charge and discharge became shorter. This caused the capacitor to not reach its maximum or minimum voltage levels before the next cycle started. As a result, the waveform across the capacitor appeared irregular or uneven. The higher the frequency, the more noticeable this irregularity became.

3.2 Task 2

1. First we constructed the circuit stated in second part of the preliminary work as shown in figure 10.

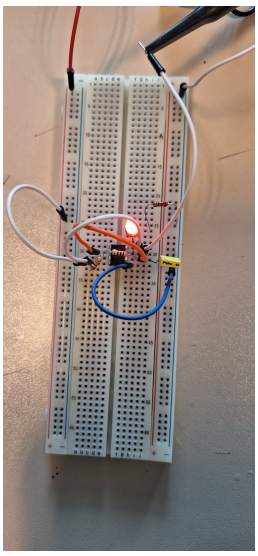


Figure 10: Physical form of circuit stated in preliminary part 2.

2. Timer circuit seen here. A LED is added to the circuit, which was initially a first order circuit. Our goal is for the LED to flash once every one second. That indicates that the LED will turn on for 0.5 seconds before turning off. For this purpose, we must select certain resistor values in order to determine the value of the capacitor. Our resistors and this capacitor will meet the requirement.

3. We will apply the formulas $f = 1.44/((R_1 + 2R_2)C)$ and $D = (R_2)/(R_1 + 2R_2)$in this task. D is assumed to be 0.5 for the first equation. In the second equation, we set the oscillopse frequency to 1 and pick the resistor values on our own. We use the resistor values of 1K and 10M ohm to make the formula above satisfy the problem.
$1.44/((1k + 20M)C) = 1$
$10^{-3}(1 + 2 * 10^{-4})C = 1.44$
$C = (144 * 10^{-5})/(1 + 2 * 10^{-4})$
$7 * 10^{-8} = 70nF$ This computation leads us to the conclusion that the capacitor is 70nF.

4. After creating circuit with 70nF and supplying 5 volt via power supply, we connected the circuit to oscilloscope.

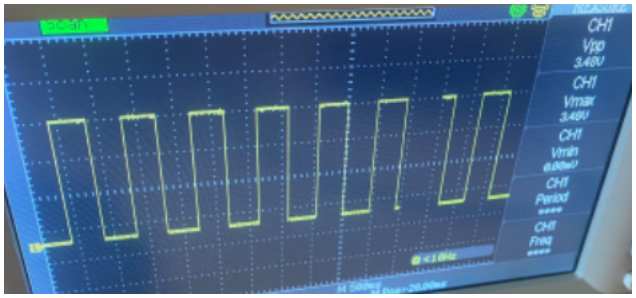


Figure 11: Oscilloscope output of circuit shown in figure 10 with given 5 Volts as input.

5. As we rate the top side and bottom side of the graph we find 1 so that shows us, the frequency is 1Hz.

4 Conclusion

We utilized the LM555 timer IC to build a circuit that blinks an LED during the experiment while observing the behavior of first-order circuits. The waveform across the capacitor changed as we increased the input signal's frequency. Higher frequencies caused the waveform to become more distorted or irregular because the capacitor had less time to charge and discharge completely before the following cycle. The circuit's time constant controls this outcome. Overall, the experiment gave helpful guidance for understanding circuit analysis and showed how to use the LM555 timer IC to successfully produce the desired LED blinking pattern.